



**American International University – Bangladesh**

**Faculty of Computer Science**

**Department of CSE**

**Course Name: Introduction to Data Science**

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**Section: D**

**Group No: 01**

**Final Term**

***Report's Title: Text Mining and NLP Analysis of BBC  
News Articles Using R.***

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1. Introduction

This project applies Natural Language Processing (NLP) and Text Mining techniques to analyze a dataset of BBC news articles. The aim is to clean, preprocess, and explore the text to uncover patterns, frequent words, sentiments, topic modeling. The study also demonstrates Named Entity Recognition (NER) using rule-based regex methods.

2. Dataset Description

- **Source:** BBC News dataset
- Dataset Overview

| Metric                   | Value                                           |
|--------------------------|-------------------------------------------------|
| Categories Analyzed      | Business, Health, Politics, Science, Technology |
| Total Words Processed    | 120,073                                         |
| Unique Words             | 555                                             |
| Total Number of Articles | 250                                             |
| Missing Values           | 0                                               |
| Number of Column         | 7                                               |

3. Data Preprocessing

3.1 Cleaning

- Removed rows with missing, empty, or very short text (<50 characters).
- Stripped HTML tags and unwanted symbols.
- Normalized text with whitespace handling.

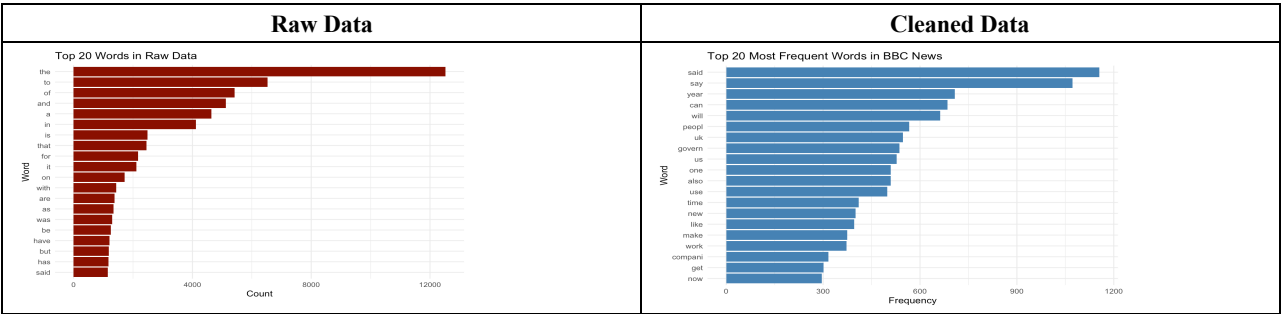
3.2 Text Preprocessing

- Converted text to lowercase.
- Removed numbers, punctuation, and English stopwords.
- Applied stemming (reducing words to their root form).
- Final cleaned text stored in a new column: Cleaned\_Content.

4. Exploratory Text Analysis

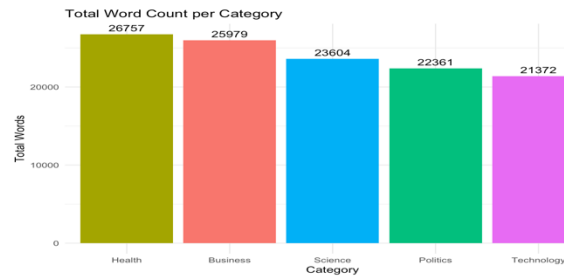
4.1 Word Frequency

- Raw word frequencies and cleaned text frequencies were computed.
- Top 20 most frequent words were visualized.



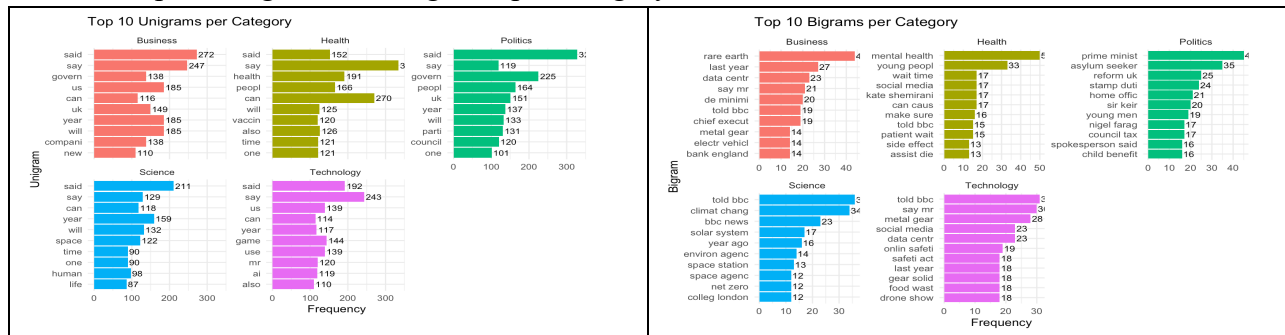
## 4.2 Category-wise Word Count

- Counted total words per category.
- Business and Health categories had higher overall word counts compared to others.



## 4.3 N-grams (Unigrams & Bigrams)

- Extracted unigrams and bigrams.
- Top 10 unigrams and bigrams per category were identified and visualized.



## 4.4 Word Clouds

- Word clouds generated for each category.
- Provided intuitive visualization of dominant terms.



Fig: Business Category

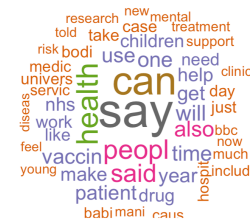


Fig: Health Category

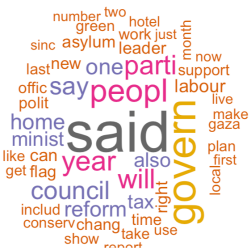


Fig: Politics Category



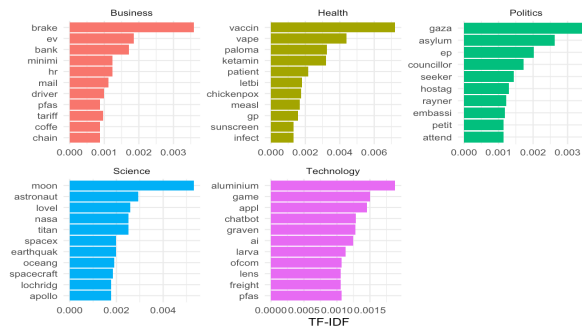
Fig: Science Category



Fig: Technology Category

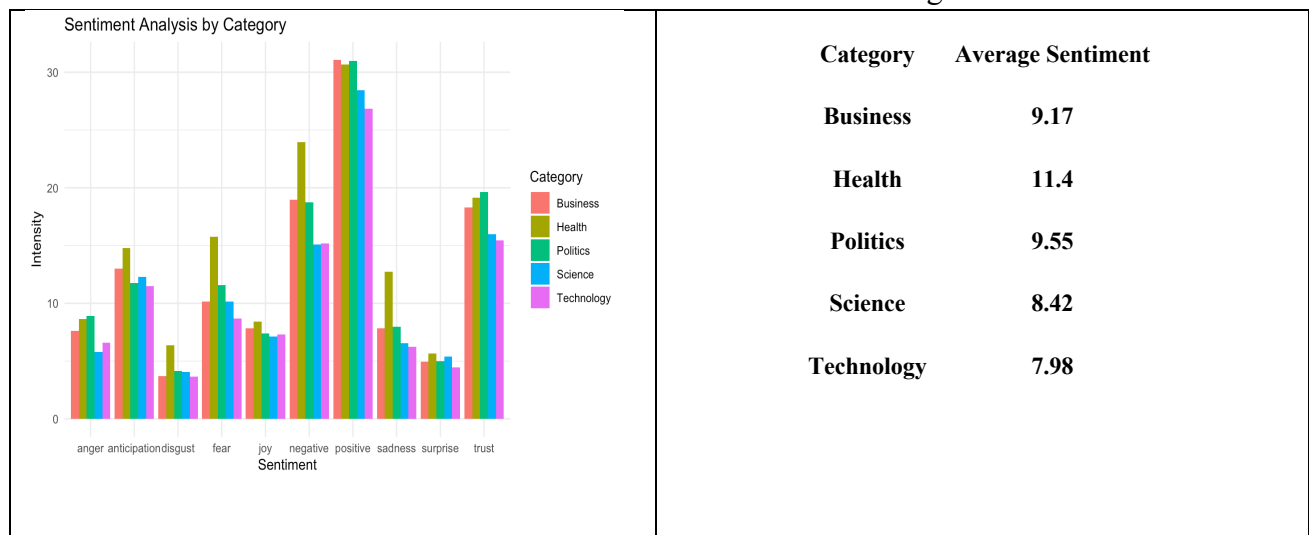
#### 4.5 TF-IDF Analysis

- TF-IDF calculated per category.
- Top distinctive terms per category identified.



#### 5. Sentiment Analysis

- **Method:** NRC Lexicon (via syuzhet package).
- Sentiments: anger, fear, joy, sadness, trust, etc.
- Average sentiment values calculated per category.
- Visualization showed differences in emotional tone across categories.



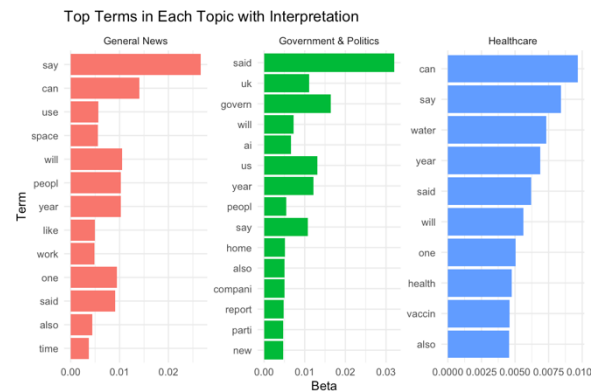
## 6. Topic Modeling

- **Method:** Latent Dirichlet Allocation (LDA).
- **Number of Topics:** 5.
- **Top Terms:** Extracted for interpretation.

### 6.1 Topic Interpretations

1. **Gaming & Entertainment** – words like *game, play, seri*.
2. **Finance & Economy** – words like *bank, tax, economi*.
3. **Healthcare** – words like *health, care, patient*.
4. **Government & Politics** – words like *govern, polit, state*.
5. **Technology & Security** – words like *tech, data, secur*.

| Topic Number | Top Terms                                      | Interpretation        |
|--------------|------------------------------------------------|-----------------------|
| 1            | say, can, peopl, use, space, will, said, work  | General News          |
| 2            | said, uk, govern, will, us, year, peopl, parti | Government & Politics |
| 3            | can, say, water, year, said, will, one, health | Healthcare            |
| 4            | said, us, ai, say, year, govern, home, also    | Government & Politics |
| 5            | say, can, year, will, like, one, also, peopl   | General News          |



## 7. Named Entity Recognition (NER)

- **Method:** Regex-based enhanced extraction.
- **Entities Extracted:** Persons, Locations, Organizations, Dates, Money.
- Tested on sample articles for validation.

### ENHANCED REGEX METHOD:

Persons: From Friday, President Donald Trump, Katherine Theobalds, Buenos Aires, Zou Xou, Joe Biden, Peter Navarro, White House, Mainland China, Hong Kong, Tam Nguyen, South East Asia, Christopher Lundell, United States, Mr Lundell, Ms Nguyen, Some American, Gap Inc, Deborah Elms, Ms Elms, Hinrich Foundation, Wool War ehouse, Andrew Smith, Mr Smith, At Zou Xou, Ms Theobalds, New York, And China, Nadine Yousif, Bernd Debusm ann Jr

Locations: The US, East Asia

Organizations: Gap Inc, Hinrich Foundation

Dates:

Money:

### ENHANCED REGEX METHOD:

Persons: Kiwi Farms, Online Safety Act, Federal Court, Preston Byrne, Community Support, Ronald Coleman, ( oleman Law Firm, Fifth Amendments

Locations: The US

Organizations:

Dates: 17 April 2025

Money:

## 9. Results Summary

- Data Preprocessing
- **Categories Analyzed:** Politics, Business, Health, Science, Technology.
- **Unique Words:** Based on word frequency.
- **Category wise word count**
- **Sentiment Analysis** – find the pattern of sentiment (anger, fear, joy, sadness, trust, etc.).  
Average sentiment values calculated per category
- **Topic Modeling** – discover hidden themes or topics within large collections of text.
- **Named Entity Recognition (NER)** – identify names of people, places, organizations.

## 10. Conclusion

This project successfully applied NLP methods to analyze BBC News articles. Insights include: - Frequent terms highlight category-specific language. - Sentiment analysis reveals tone differences across topics. - Topic modeling clusters articles into interpretable groups. - Regex-based NER identified key entities.

The workflow establishes a comprehensive pipeline for text analytics, applicable to news datasets and beyond.

## 11. Future Work

- Improve NER using deep learning models (e.g., spaCy, BERT).
- Extend topic modeling with more advanced methods like BERTopic.
- Perform classification to predict article category from text.
- Time-based sentiment and topic trend analysis.